Generative AI
(23CA301PC202)

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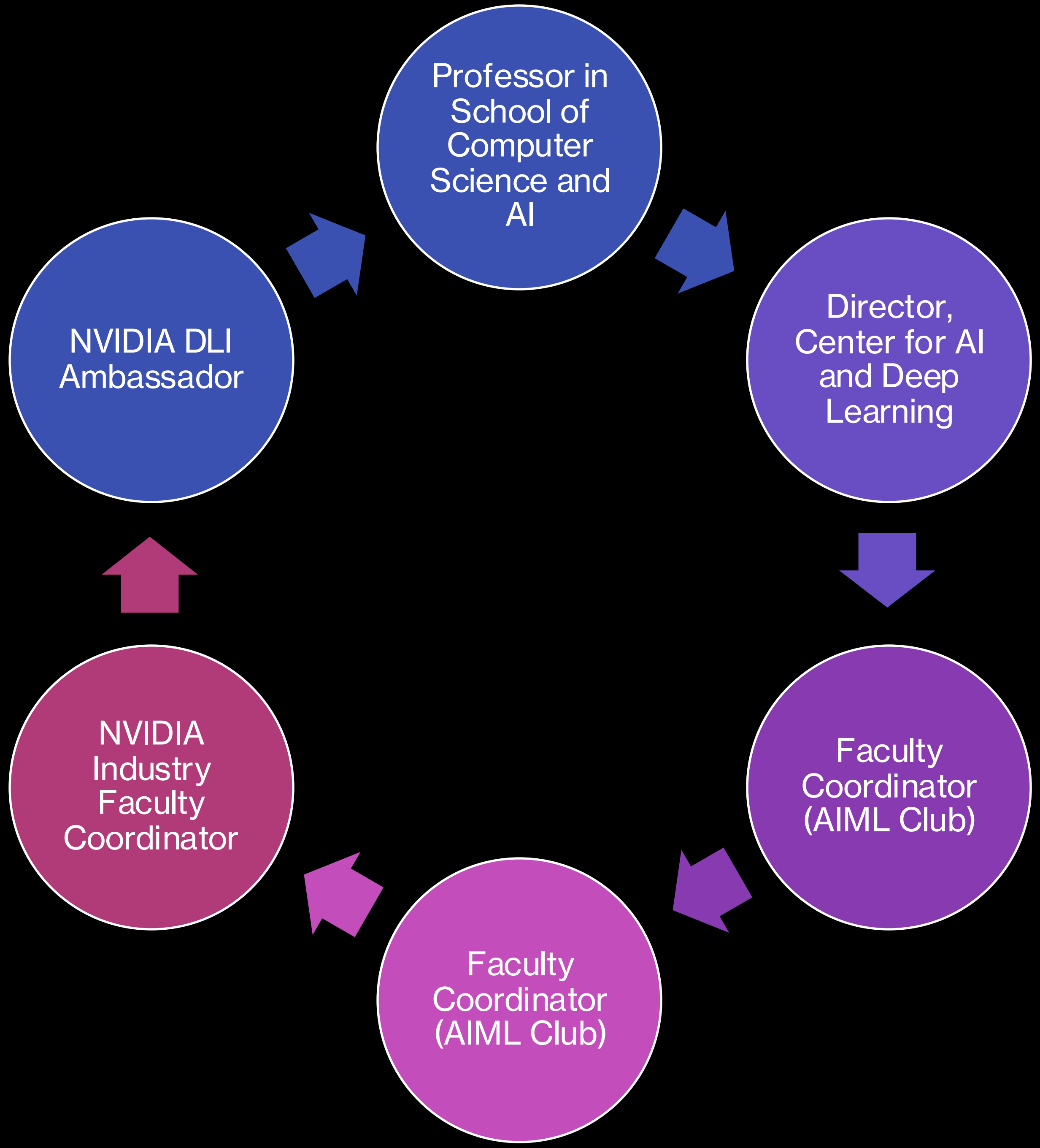

Generative AI (UNIT-1)

- Introduction to Generative AI: Applications of Generative AI in various fields
- Fundamentals of Neural Networks: Perceptron and Multilayer Perceptron (MLP)
- Gradient Descent, Back propagation Algorithm, Loss functions, Understanding Bias and Variance

Generative AI (UNIT-2: Computer Vision)

- Introduction to CNN
- Convolution neural networks (CNNs)
- Understanding the basics of CNN
- Transfer Learning
- Fine-tuning

Generative AI (UNIT-3: Image Generation using Generative Models)

- Image Generation using Generative Models
- Variational Autoencoders (VAE)
- Generative Adversarial Networks (GANs)
- Conditional Generative Models
- Advantages and Limitations of VAEs and GANs

Generative AI (UNIT-4: Sequential Data Generation using Generative Models)

Recurrent Neural Networks (RNNs)

Applications of RNNs in Generative AI
- Text generation with RNN
- Music generation using RNN

Long short-term memory (LSTM)

Applications of LSTM in Generative AI
- Text generation with LSTM
- Music generation using LSTM

Generative AI
(UNIT-5: Recent Trends)

- Attention mechanism
- Transformer Models
- Diffusion Models

Text Books

1

“Generative Deep Learning” by David Foster published by O'Reilly Media 2nd edition 2023

2

“Generative AI with LangChain” by Ben Auffarth published by Packt Publishing, 2023

Evaluation Pattern

- Mid Term Examination : 20 Marks
- End Term Examination : 40 Marks
- Lab Exam (2 tests) : 5 X 2 = 10
  - First Exam: In First Week, After Completing 5 Lab Assignments
  - First Exam: In First Week, After Completing 10 Lab Assignments
- Certification : 5 Marks
- Project : 15
  - Review 1 : 5
  - Review 2 : 5
  - Report/Paper : 5
- Class Tests (2 tests) : 5 X 2 = 10
  - First Test: After Completion of 1st unit
  - Second Test: After Completion of 3rd unit

MOOC Courses For Certification

1. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01380915637902 5408139_shared/overview
2. https://www.coursera.org/learn/generative-ai-prompt-engineering-for-everyone
3. https://www.coursera.org/learn/generative-ai-introduction-and-applications
4. https://www.coursera.org/learn/generative-ai-for-everyone
5. https://www.coursera.org/specializations/generative-ai-for-everyone
6. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_0138091787684 16768124_shared/overview

What is generative AI?

- Generative AI is artificial intelligence designed to create unique text or image results in response to user prompts.
- The technology uses deep learning to return an output based on the user's prompt.
- Generative AI is used to create new written, visual, or audio content, summarize complex data, generate code etc.
- AI engineers train the technology using large data sets, which the model consults when determining the best possible answer to a prompt.

Generative AI: Tools

ChatGPT or DALL-E: Generative artificial intelligence created by OpenAI, a Microsoft-backed, profit-capped company with the mission to develop artificial intelligence to serve humankind (https://openai.com/index/dall-e-2/)

Google Bard: Google's generative AI with integrations to Google products like Google Lens and Gmail, operating with a language model called PaLM-2 that was trained on the largest data set out of all generative AI models available at the time of its release(https://gemini.google.com/app?hl=en-IN)

Applications of generative AI

HEALTH CARE AND PHARMACEUTICALS

ADVERTISING AND MARKETING

MANUFACTURING

SOFTWARE DEVELOPMENT

FINANCIAL SERVICES

MEDIA AND ENTERTAINMENT

EDUCATION

Health care and pharmaceuticals

Enhancing medical images: Generative AI can augment medical images like X-rays or MRIs, synthesize images, reconstruct images, or create reports about images. This technology can even generate new images to demonstrate how a disease may progress in time.

Discovering new drugs: Researchers can use generative AI to research and develop new medicines. Gartner projects that 30 percent of the new drugs created by researchers in 2025 will use generative design principles

Simplify tasks with patient notes and information: Generational AI can build patient information summaries, create transcripts of verbally recorded notes, or find essential details in medical records more effectively than human efforts.

Personalized treatment: Generative AI can consider a large amount of patient information, including medical images and genetic testing, to deliver a customized treatment plan tailored to the patient's needs.

Advertising and marketing

Generate marketing text and images: Generative AI can help marketing professionals create consistent, on-brand text and images to use in marketing campaigns. This technology also offers translation tools to spread your marketing message into new territories. Gartner predicts that marketing professionals will use generative AI to create 30 percent of outbound marketing materials by 2025

Generate personalized recommendations: Generative AI helps create powerful recommendation engines to help customers discover new products they might like. With generative AI, this process is more interactive for customers.

Create product descriptions: Beyond flashy advertising campaigns, generative artificial intelligence can help with tedious or time-consuming content requirements like creating product descriptions.

Manufacturing

Accelerating the design process: Using generative AI, engineers and project managers can work through the design process much faster by generating design ideas and asking the AI to assess ideas based on the constraints of the project.

Provide smart maintenance solutions for equipment: Maintenance professionals can use generative AI to track the performance of heavy equipment based on historical data, potentially alerting them to trouble before the machine malfunctions. Generative AI can also recommend routine maintenance schedules.

Software development

Generating code: Software developers can create, optimize, and auto-complete code with generative AI. Generative AI can create code blocks by comparing them to a library of similar information. It can also predict the rest of the code

Translate programming languages: Generative AI can be a tool for developers to interact with software without needing a programming language. The generative AI would act as a translator.

Financial services

Create	Create investment strategies: Generative AI can recommend the best investments according to your or your client's goals. This technology can find and execute trades much faster than human investors
Communicate and educate	Communicate and educate clients and investors: Financial services professionals sometimes need to communicate complex information to clients and colleagues.
Draft	Quickly draft documentation and monitor regulation: Generative AI can monitor regulatory activity, keep you informed of any changes, and create drafts of documents such as investment research or insurance policies.

Media and entertainment

Create audio and visual content: Generative AI can create new video content from scratch. This tech can also help you make visual content faster by creating visual effects, adding graphics

Generate highlights for sports and events: When it comes to sporting and live events, gen AI can create highlight reels instantly and allow fans to create their own custom highlights. For example, fans could generate highlights of a particular play or a tournament series.

Education

- Data insights on student performance
- Assignments creation
- Generation of PPT based on contents
- Emails drafts
- Questioner preparation

Thank You

Reference:
https://www.coursera.org/articles/generative-ai-applications